

Lecture Homework Week 07 - Thursday Template

Student Information

Name: Laiba Nasir

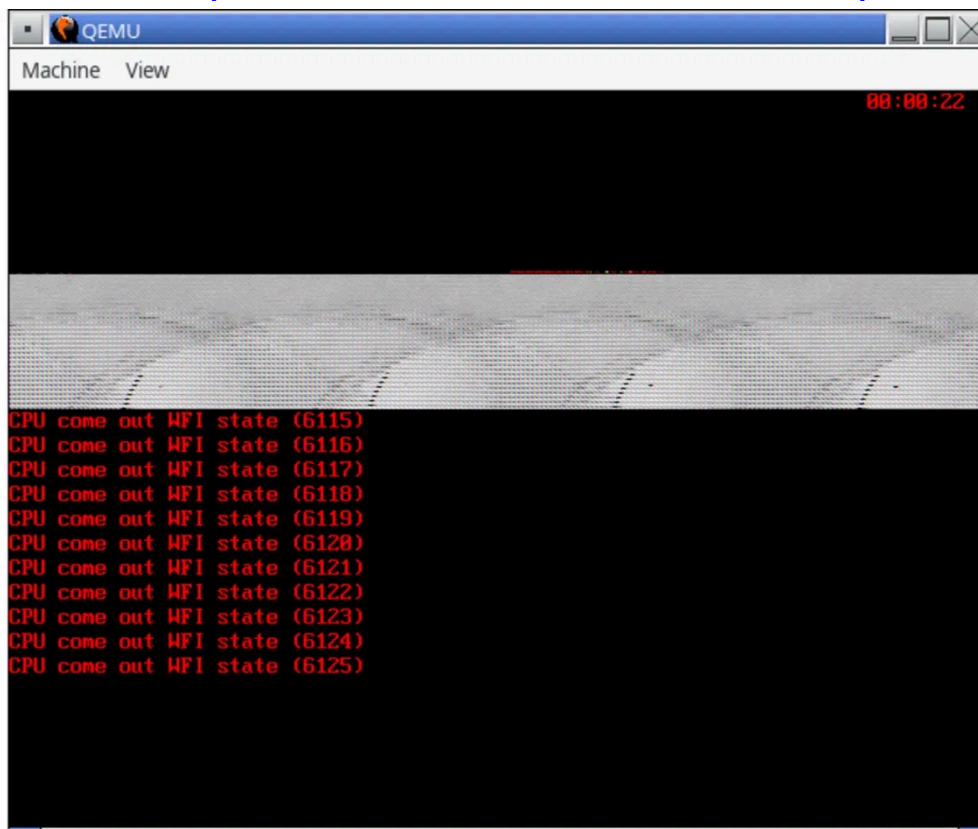
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Student Responses

Answer the following questions

Question 1 Comment out the asm line. Run the program again to see what would happen. What do you observe that happens differently than before:

When I comment out the ASM line and run the program again, I notice the timer slows down and the messages are super fast. So in the screenshot, it says 00:22 but the loop counter is already 6125. This means the CPU looped over 6000 times in just 22 seconds instead of once per second. This shows us the CPU never sleeps.



The screenshot shows a QEMU window titled "Machine View". In the top right corner, a red timer displays "00:00:22". The main area of the window shows a list of messages in red text, indicating the CPU coming out of the WFI (Wait For Interrupt) state at various loop counter values:

```
CPU come out WFI state (6115)
CPU come out WFI state (6116)
CPU come out WFI state (6117)
CPU come out WFI state (6118)
CPU come out WFI state (6119)
CPU come out WFI state (6120)
CPU come out WFI state (6121)
CPU come out WFI state (6122)
CPU come out WFI state (6123)
CPU come out WFI state (6124)
CPU come out WFI state (6125)
```

Question 2 In terms of CPU power consumption, what difference does the asm statement make?

The ASM makes a difference because when we do not comment it out, the CPU is in a low power state, and only wakes up when an interruption occurs. This causes us to use less power. However, when we comment out ASM, the CPU runs at full speed at constant time,

and what ends up happening is wasting power when there's no need because there's nothing useful going on. In the picture, for embedded systems in general, this difference makes a big impact on the battery life of Embedded devices; it changes the battery life from going days versus hours